

## Ques. HOW TO OPERATE 60HZ ELECTRICAL APPLIANCES AT 50HZ?

**Momtaz**

**Ans.** First of all try to contact the manufacturer of the appliance for safely operating at 50Hz with the manufacturer's approval. The manufacturer may have a solution.

It is possible to purchase 60Hz sine-wave power systems that take 240V 50Hz on input and produce 115V 60Hz on output. These solutions are generally very expensive, and it is usually more cost-effective to replace the tool or appliance.

Visicomm Industries manufactures, sells and rents a broad range of rotary and solid-state frequency converters and changer products rated from 1kVA to 5000kVA.

A frequency converter can be for single or three phases and can sometimes also change voltage, thus functioning as a power converter from 60Hz to 50Hz, from 50Hz to 60Hz and from 50Hz/60Hz to 400Hz. It can also be used for continuous duty or for testing products designed for export.

## Q2. WHAT ARE THE CONSIDERATIONS IN SMPS DESIGN? PLEASE PROVIDE LINKS OR E-BOOKS FOR THE SAME.

**Mahesh Kumar**

**A2.** In an SMPS design, you have to consider power output, input conditions, thermal and heat-sinking issues, component quality, PCB area, layout and footprint, performance testing and much more. Please follow the link for more details: [electronicdesign.com/power/7-critical-steps-switching-power-supply-design](http://electronicdesign.com/power/7-critical-steps-switching-power-supply-design)

You may refer to the following SMPS books: *Switch-Mode Power Sup-*

*plies* by C.P. Basso, *Switching Power Supply Design* by A.I. Pressman, and *Modern DC-To-DC Switchmode Power Converter Circuits* by R.P. Severns and G.E. Bloom.

You can also find a complete e-guide on SMPS design from the following link: [http://caxapa.ru/thumbs/348441/Switchmode-Power-Supply-Handbook\\_3rd\\_ed.pdf](http://caxapa.ru/thumbs/348441/Switchmode-Power-Supply-Handbook_3rd_ed.pdf) -- handbook and [http://www.smtps.com/Knowledge/Articles/Step-by-Step-Fly-back\\_SMPS\\_Design.shtml](http://www.smtps.com/Knowledge/Articles/Step-by-Step-Fly-back_SMPS_Design.shtml)

You can design an SMPS circuit based on the following online simulators freely available: [www.powersim.com](http://www.powersim.com) and [www.ti.com/lscds/ti/analog/webench/power.page](http://www.ti.com/lscds/ti/analog/webench/power.page)

## Q3. WHAT TYPES OF ACCELEROMETER SENSORS ARE AVAILABLE IN THE MARKET? WHAT POINTS SHOULD BE CONSIDERED WHEN BUYING AN ACCELEROMETER?

**Pamarthi Kanakaraja**

**A3.** There are various types of accelerometers available in the market based on various sensing principles. These are capacitive, piezoelectric, piezoresistive, Hall effect, magnetoresistive and heat transfer, etc.

Consider the following points when buying an accelerometer:

**Analogue versus digital.** This is determined by the hardware that is being interfaced with the accelerometer. Analogue style accelerometers output a continuous voltage that is proportional to acceleration, whereas digital accelerometers usually use pulse width modulation (PWM) for output.

**Number of axes.** For most projects, two axes are enough. However, if you want to attempt 3D positioning, you will need a 3-axis accelerometer.

**Maximum swing.** If you only care about measuring tilt using Earth's grav-

ity, a  $\pm 1.5g$  accelerometer will be more than enough. If you are going to use the accelerometer to measure the motion of a car, plane or robot,  $\pm 2g$  should give you enough headroom to work with. For a project that experiences very sudden starts or stops, you will need one that can handle  $\pm 5g$  or more.

**Sensitivity.** With more sensitivity, you will get more accurate readings.

**Bandwidth.** For slow-moving tilt-sensing applications, a bandwidth of 50Hz will probably suffice. If you intend to measure vibrations or control a fast-moving machine, you will need a bandwidth of several hundred Hz.

**Impedance/buffering issues.** Mainly involving analogue accelerometers, use a low-input offset rail to rail op-amp as a buffer to lower output impedance.

## Q4. HOW CAN ONE DEVELOP A DIGITAL SPEEDOMETER USING A SENSOR ATTACHED TO THE FRONT OR REAR WHEEL THAT ALSO DISPLAYS DISTANCE (KM) AND SPEED (KMPH)?

**Aman Madan**

**A4.** Please refer 'Microcontroller Based Speedometer-Cum-Odometer' article published in EFY magazine in November 2008 issue. In the article you will find the following features: digital readout, speed displayed in kmph, distance travelled displayed in kilometers, readings saved in non-volatile memory (EEPROM), home-brewed speed transducer/sensor, self-reset to zero after completion of 99,999.9km and easy-to-build-and-fix onto the bike.

Answers compiled by EFY senior application engineer, Nidhi Kathuria. Letters and questions for publication may be addressed to Editor, Electronics For You, D-97/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: [editor@efy.in](mailto:editor@efy.in)) and should include name and address of the sender